Higher National Diploma in Engineering (Electrical)

First Year, 1St Semester Examination — 2022

EE 1107 Electrical Principals A

Question 1

(a) How does the increase in temperature effect the resistance of,

I. Pure metals

II. Alloys

III. Insulators (03 Marks)

(b) Obtain the resultant resistance across AB for the circuit shown in figure 1(b) below. If 12V DC supply is given, what would be the current through the resultant resistance? (05 marks)

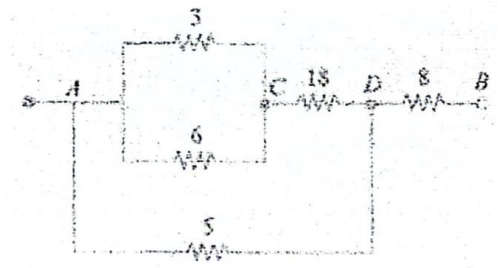


Figure – 1(b)

(c) Identify the value of resistor with tolerance from their colour codes as shown in below Figure-1(c).(04 Marks)

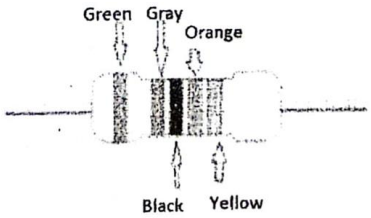


Figure – 1(c)

(d) Determine the currents in AB, CD and DF branches of the circuit in Figure - 1(d) by using Kirchhoff's Laws.(08 marks)

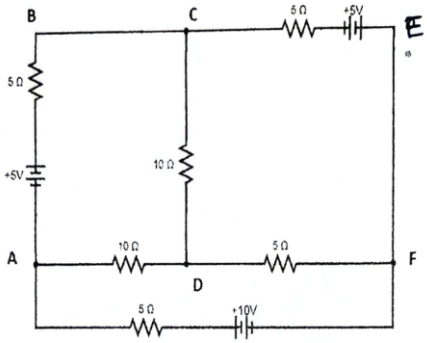


Figure – 1(d)

Question 2

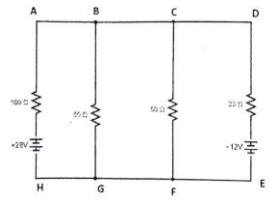
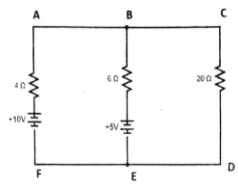


Figure – 2(a)

a) Using Milliman Theorem, find currents in AH, BG, CF and DE branches in Figure -2(a).

(08 marks)



b) Using Thevenin's Theorem, find the current across 20Ω resistor in Figure 1(b) above.(06 marks)

c) Draw the transient response of RC circuit for charging and discharging clearly showing the transient and steady states.(04 marks)

d) Give two real life applications of RC circuits.(02 marks)

Question 3

(a) State **Coulomb's law**. (02 marks)

(b) Two-point charges, of $8\mu\mathrm{C}$ and $4\mu\mathrm{C}$ are separated by a distance of $15\ \mathrm{cm}$. What is the magnitude of the electric force between them? (04 marks)

(c) Give three properties of **electric field lines**. (03 marks)

(d) What is known as an **electric breakdown**? (02 marks)

(e) Name three types of **capacitors**. (03 marks)

(f) A capacitor having a capacitance of $80\mu\mathrm{F}$ is connected across a $500\ \mathrm{V}\ \mathrm{d.c.}$ supply. Calculate the **Electric charge** in the capacitor. (02 marks)

(g) A circular coil of radius $5 \times 10^{-2}\ \mathrm{m}$ with $40\ \mathrm{turns}$ is carrying a current of $0.25\ \mathrm{A}$. Using **Bio Savart Law**, determine the **magnetic field** of the circular coil at its center. (04 marks)

Question 4

(a) State the two **Faraday's laws of electromagnetic induction**. (04 marks)

(b) What are the four factors affecting the **inductance** of a circuit? (04 marks)

(c) What is the difference between **self-inductance** and **mutual inductance**? (02 marks)

(d) A conductor having $200\ \mathrm{mm}$ long carries a current of $10\ \mathrm{A}$ and is at right angles to a magnetic field lying between two circular pole faces each of radius $50\ \mathrm{mm}$. If the total flux between the pole face is $0.5\ \mathrm{mWb}$, Calculate the magnitude of **force exerted on the conductor**. (05 marks)

(e) What is known as **coefficient of coupling**? (02 marks)

(f) Determine the **magnetic field strength** required to produce a flux density of $0.25\ \mathrm{T}$ in an air gap of length $10\ \mathrm{mm}$. (03 marks)

Question 5

(a) Explain how an **ideal diode** acts as a switch. (04 marks)

(b) Draw the V-I **characteristic curve** for a real Si diode by indicating key points on the curve. (04 marks)

(c) What is a **half-wave rectifier**? (02 marks)

(d) Draw the diagram of a **full wave bridge rectifier** and clearly shows the input and output voltage locations. (04 marks)

(e) Explain the **operation of full wave rectifier circuit**. (04 marks)

(f) What is the use of a **Zener diode**? (02 marks)

Question 6

(a) Briefly explain the difference between a **clipper circuit** and a **clamper circuit** by illustrating with sketches. (04 marks)

(b) Draw the circuit and expected output from a **positive clamper circuit**. (04 marks)

(c) Draw the **symbol diagram** for a $\mathrm{n\text{-}p\text{-}n}$ transistor. (01 mark)

(d) Explain how a $\mathrm{n\text{-}p\text{-}n}$ transistor is **doped**. (03 marks)

(e) A bipolar $\mathrm{n\text{-}p\text{-}n}$ transistor has a DC current gain (β) value of 200. Calculate the minimum base current I_B required to switch a resistive load, carrying $4\ \mathrm{mA}$ current. (03 marks)

(f) Draw the three transistor configurations (**Common base**, **Common collector**, **Common emitter**) for a $\mathrm{p\text{-}n\text{-}p}$ transistor showing input and output voltages. (03 marks)

(g) What are the two main areas of **transistor usage** in day-to-day applications? (02 marks)